Year Up United
Data Analyst Training Academy: Intro to Data Analysis & Excel Fundamentals

Week 1 Student Guide

This courseware is to be distributed to accompany instructor led training.

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Table of Contents

MODULE 1 GETTING STARTED 5

SECTION 1.A INTRODUCTION 6

1.A.1 Course Introduction 7
1.A.2 Best Practices for Success: Practical 8
1.A.3 Best Practices for Success: Mindset 9
1.A.4 Useful resources 10
1.A.5 A caution about Gen AI tools 11
1.A.6 Week 1 Learning Objectives 12
1.A.7 Week 1 Learning Outcomes 13

MODULE 2 DATA ANALYSIS BASICS 14

SECTION 2.A DATA FUNDAMENTALS 15

2.A.1 What is data? 16
2.A.2 Quantitative vs. categorical 17
2.A.3 Structured vs. unstructured data 18
2.A.4 Common sources of data 19
2.A.5 Common formats for data storage 20
2.A.6 Data is fallible 21
2.A.7 Reading data tables 22
2.A.8 Tabular formats 23
2.A.9 What is data analysis? 24
2.A.10 Data analysis roles 25
2.A.11 Key concepts for data analysis 26
2.A.12 Math fundamentals 27
2.A.13 Statistics concepts 28
2.A.14 More statistics concepts 29
2.A.15 Correlation and causation 30
2.A.16 Data at work: Business Intelligence 31
2.A.17 Data security and data governance 32

SECTION 2.B DATA ANALYSIS LIFECYCLE 33

2.B.1 Types of data analysis 34
2.B.2 The data analysis process 35
2.B.3 Question presented 36
2.B.4 Data discovery and acquisition 37
2.B.5 Clean and transform 38
2.B.6 Data Modeling 39
2.B.7 Analyze 40
2.B.8 Interpret 41
2.B.9 Share results 42

MODULE 3 GATHERING DATA 43

SECTION 3.A DATA COLLECTION 44

3.A.1 Collecting data 45
3.A.2 Good data collection practices 46
3.A.3 Common pitfalls in data collection 47

3.A.4 Collecting data with surveys *48*
3.A.5 Survey basics *49*
3.A.6 Survey best practices *50*
SECTION 3.B PUBLIC DATA SOURCES 51
3.B.1 Publicly available data sources *52*
3.B.2 More searchable public data sets *53*
3.B.3 Common issues with public data *54*
SECTION 3.C BIAS IN DATA 55
3.C.1 Bias in data collection and analysis *56*
3.C.2 Cognitive bias vs. cultural bias *57*
3.C.3 How can data be biased? *58*
3.C.4 Selection bias *59*
3.C.5 Noticing and addressing bias *60*
3.C.6 How to talk about bias in data *61*

MODULE 4 EXCEL FUNDAMENTALS 62

SECTION 4.A EXCEL BASICS 63
4.A.1 What is Microsoft Excel? *64*
4.A.2 Navigating the Excel interface *65*
4.A.3 Entering and Reviewing Data *66*
4.A.4 Using the Ribbon *67*
4.A.5 Columns and Rows *68*
4.A.6 Working with Columns and Rows *69*
4.A.7 Ranges *70*
4.A.8 Drag to Fill and AutoFill *71*
4.A.9 Formula Basics *72*
4.A.10 Sorting and Filtering *73*
SECTION 4.B CLEANING & RESHAPING DATA WITH EXCEL 74
4.B.1 Data Types and Formatting *75*
4.B.2 Modifying Data Types *76*
4.B.3 Transposing Columns and Rows *77*
4.B.4 Removing Duplicates *78*
4.B.5 Cautions for Removing Duplicates *79*
4.B.6 Splitting Cell Text to Columns *80*
4.B.7 Using the Text to Columns Wizard *81*
4.B.8 Find and Replace *82*
4.B.9 Using Find and Replace *83*
4.B.10 Helpful Functions for Data Cleanup *84*
4.B.11 Using IF and SWITCH *85*
4.B.12 Using VLOOKUP *86*
4.B.13 Using XLOOKUP *87*
4.B.14 Using Filters for Data Review *88*
SECTION 4.C ANALYZING AND VISUALIZING WITH EXCEL 89
4.C.1 What are visualizations? *90*
4.C.2 Creating Charts in Excel *91*
4.C.3 When to Use: Bar and Line Charts *92*
4.C.4 When to Use: Pie and Scatter Charts *93*
4.C.5 Chart Formatting Options *94*
4.C.6 More on Chart Formatting *95*
4.C.7 PivotTable Fundamentals *96*
4.C.8 Building Your PivotTable *97*

4.C.9 Pivot Charts *98*

MODULE 5 DATA STORYTELLING 99

SECTION 5.A INTRO TO DATA STORYTELLING 100

5.A.1 Introduction to data storytelling *101*
5.A.2 Why does data need storytelling? *102*
5.A.3 Understanding your audience *103*
5.A.4 Structuring your data story *104*
5.A.5 Presentation best practices *105*
5.A.6 Common Pitfalls *106*
5.A.7 Give Recommended Actions / Next Steps *107*

Module 1

Getting Started

Section 1.A

Introduction

1.A.1 Course Introduction

- Welcome! Over the next 12 weeks, we will be exploring:
  - ◊ Foundations and fundamentals of data analysis and data storytelling
  - ◊ The basics of using Excel to review and analyze data
  - ◊ Writing SQL queries and working with databases in MySQL Workbench
  - ◊ Python language basics, including reading/writing files with Python
  - ◊ Python's NumPy, Pandas, and Matplotlib modules
  - ◊ Using Microsoft Power BI for creating semantic models, reports, and dashboards
- Your participation and questions are critically important in making the most of this experience
  - ◊ There will be lots of different opportunities to engage: class discussions, breakout groups, and asynchronous chat
  - ◊ **All questions are good questions** – sometimes even a very simple question can change your whole understanding of things
- This material tries very hard not to make any assumptions about what you might already know
  - ◊ If you feel like you already do know some of this stuff, that's great! Please chime in with your own examples and experiences
  - ◊ On the flip side, if there are words or tools or concepts that come up in discussion which are not familiar to you, please raise the question!

1.A.2 Best Practices for Success: Practical

- Develop good work habits:
  - ◊ Be on time and ready to get started each morning
  - ◊ Take notes in a consistent place so that you can find them back again
  - ◊ Take breaks when you need to, and ask questions when you're stuck
- Set up a comfortable workspace:
  - ◊ Use a second monitor
  - ◊ Use an external keyboard that you find comfortable to type on
    - ∗ If you're a "hunt and peck" typist, practice touch-typing for 10-30 minutes a day (typingclub.com or keybr.com are good options)
  - ◊ Use an external mouse that you can place within easy distance of your keyboard
  - ◊ Have a little free desk space to keep a notebook / scrap paper and pen, plus any other items you need to work comfortably
    - ∗ But ideally, put your phone face down and out of reach
  - ◊ Arrange your computer, second monitor, and keyboard/mouse so that your workspace is ergonomically friendly:
    - ∗ You're able to sit comfortably upright, ideally with the top of your computer screen at about eye level
    - ∗ Your arms are at a comfortable 90 degree angle when you type, without scrunching up your shoulders

1.A.3 Best Practices for Success: Mindset

- **Curiosity** is your most powerful tool as an analyst
  - ◊ Ask questions! People you work with will typically be happy to share their knowledge and experience, and you will show up as a person who is interested in how things work
  - ◊ Break stuff to figure how it works – this is what test environments are for!
  - ◊ Be open to feedback
    - * This means focusing less on "what I meant" and more on "what the other person thought I meant"
- Sometimes learning from your mistakes is part of the process
  - ◊ "The master has failed more times than the apprentice has tried"
- Keep an open mind – there is always more to learn!
  - ◊ In the world of knowledge, there are:
    (1) things you know
    (2) things you know that you don't know
    (3) things you don't know that you don't know
  - ◊ Half the battle is moving from (3) to (2), so that you can get from (2) to (1)
- Be kind to yourself
  - ◊ Think about someone who you love and respect – how would you treat them when they are trying to learn something new? Try to treat yourself with the same supportiveness and courtesy

1.A.4 Useful resources

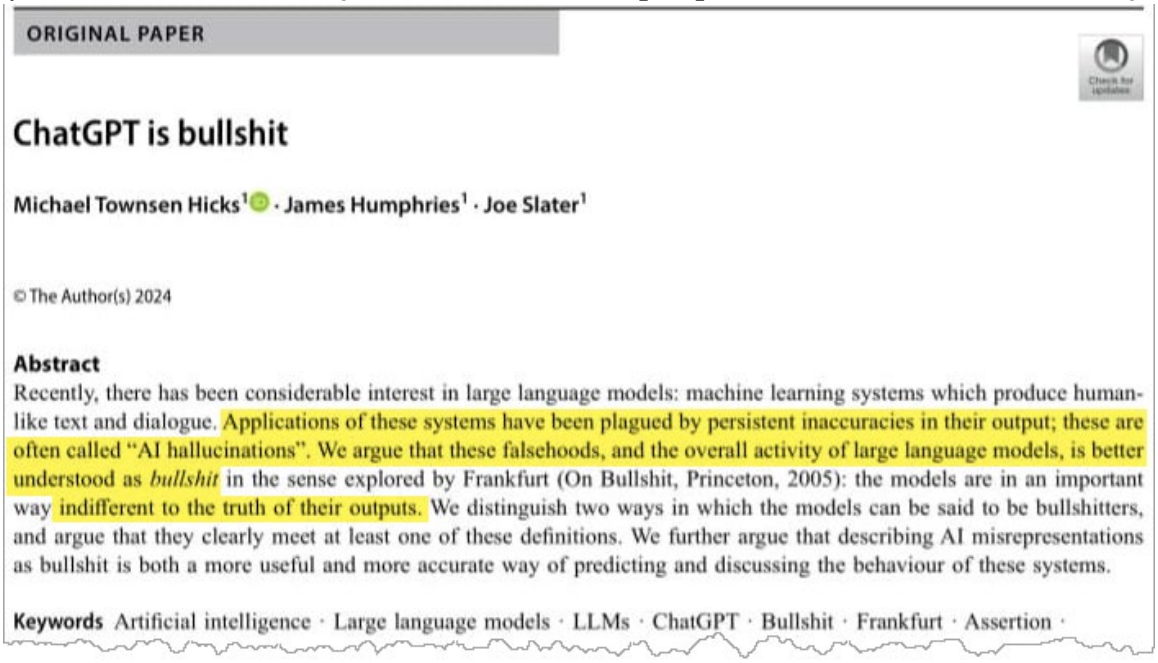
- In addition to in-class resources, you are welcome to search the web for information about any of the topics we discuss – just be thoughtful about where the results are coming from
  - ◊ For example, discussions on Stack Overflow can be really helpful – but pay attention to how old the discussion thread is, because some information may be outdated
  - ◊ Sometimes you can find a good YouTube video walking through how to solve a particular problem – but again, pay attention to age of videos
- The best approach is to **try it yourself first**, breaking down the problem to take it one little piece at a time
- For helpful guidance:
  - ◊ W3Schools (www.w3schools.com) has great resources on SQL generally, MySQL, Python, NumPy, Pandas, and even Data Science concepts
  - ◊ Microsoft Learn (learn.microsoft.com/en-us) has useful libraries and tutorials – this also includes general topics like data analytics concepts and Python, but especially will come in handy when we get to Power BI
  - ◊ Wikipedia can be a good high-level resource for programming concepts, but it doesn't always have practical examples, and sometimes will take you down a rabbit hole

1.A.5 Using Generative AI Tools

- ChatGPT, Claude, Copilot and similar AI tools can be extremely useful, but you also have to understand their limitations

- In pre-work you practiced using these tools productively: clarifying concepts, generating practice data, and reviewing your own attempts. That's the right approach!

- Nevertheless, all of these tools have well-documented problems with “AI hallucinations”
  - ◊ (or as Hicks, et al, put it “indifferen[ce] to the truth of their outputs”)



ORIGINAL PAPER

ChatGPT is bullshit

Michael Townsen Hicks[1] · James Humphries[1] · Joe Slater[1]

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Abstract

Recently, there has been considerable interest in large language models: machine learning systems which produce human-like text and dialogue. Applications of these systems have been plagued by persistent inaccuracies in their output; these are often called “AI hallucinations”. We argue that these falsehoods, and the overall activity of large language models, is better understood as *bullshit* in the sense explored by Frankfurt (On Bullshit, Princeton, 2005): the models are in an important way indifferent to the truth of their outputs. We distinguish two ways in which the models can be said to be bullshitters, and argue that they clearly meet at least one of these definitions. We further argue that describing AI misrepresentations as bullshit is both a more useful and more accurate way of predicting and discussing the behaviour of these systems.

Keywords Artificial intelligence · Large language models · LLMs · ChatGPT · Bullshit · Frankfurt · Assertion ·

- GenAI is best used where you can **confidently assess the quality** of its work, or to help point you in a **starting direction** on a question
  - ◊ But don’t rely on it to provide answers *for* you … because it will sometimes give you information that’s out of context, unnecessarily complex, outdated by a year or more, or just made up

1.A.6 Week 1 Learning Objectives

- In pre-work, you set up Git, explored Excel and statistics concepts, and practiced working with AI tools, and we'll continue building on that foundation all of this week

- This week you will learn:
  - ◊ The nature of data: types, formats, sources, and common storage conventions
  - ◊ Fundamentals of data analysis: key concepts, roles, and relevant math and statistics
  - ◊ The data analysis lifecycle as a repeatable framework for approaching analysis problems
  - ◊ Methods and best practices for data collection, including surveys
  - ◊ Publicly available data sources and how to evaluate them
  - ◊ Bias in data: how it arises, how to recognize it, and how to address it
  - ◊ Microsoft Excel interface and core operations for working with data
  - ◊ Data cleaning and reshaping techniques in Excel
  - ◊ Data analysis and visualization using Excel charts, PivotTables, and PivotCharts
  - ◊ Principles of data storytelling and communicating findings to an audience

1.A.7 Week 1 Learning Outcomes

- By the end of the week, you will be able to:
  - ◊ Distinguish between quantitative and categorical data, and between structured and unstructured data
  - ◊ Describe the stages of the data analysis lifecycle and explain the purpose of each stage
  - ◊ Design a basic survey following established best practices for data collection
  - ◊ Identify potential sources of bias in a data set or collection method and articulate their impact
  - ◊ Navigate the Excel interface and perform core operations: entering data, sorting, filtering, and applying basic formulas
  - ◊ Clean and reshape data in Excel using tools such as Text to Columns, Remove Duplicates, Find & Replace, and functions including IF, VLOOKUP, and XLOOKUP
  - ◊ Create and format bar, line, pie, and scatter charts appropriate to the data and the question being answered
  - ◊ Build a PivotTable and PivotChart to summarize and visualize a data set
  - ◊ Structure a data finding as a narrative appropriate to a target audience, incorporating a clear recommendation

Module 2

Data Analysis Basics

Section 2.A

Data Fundamentals

2.A.1 What is data?

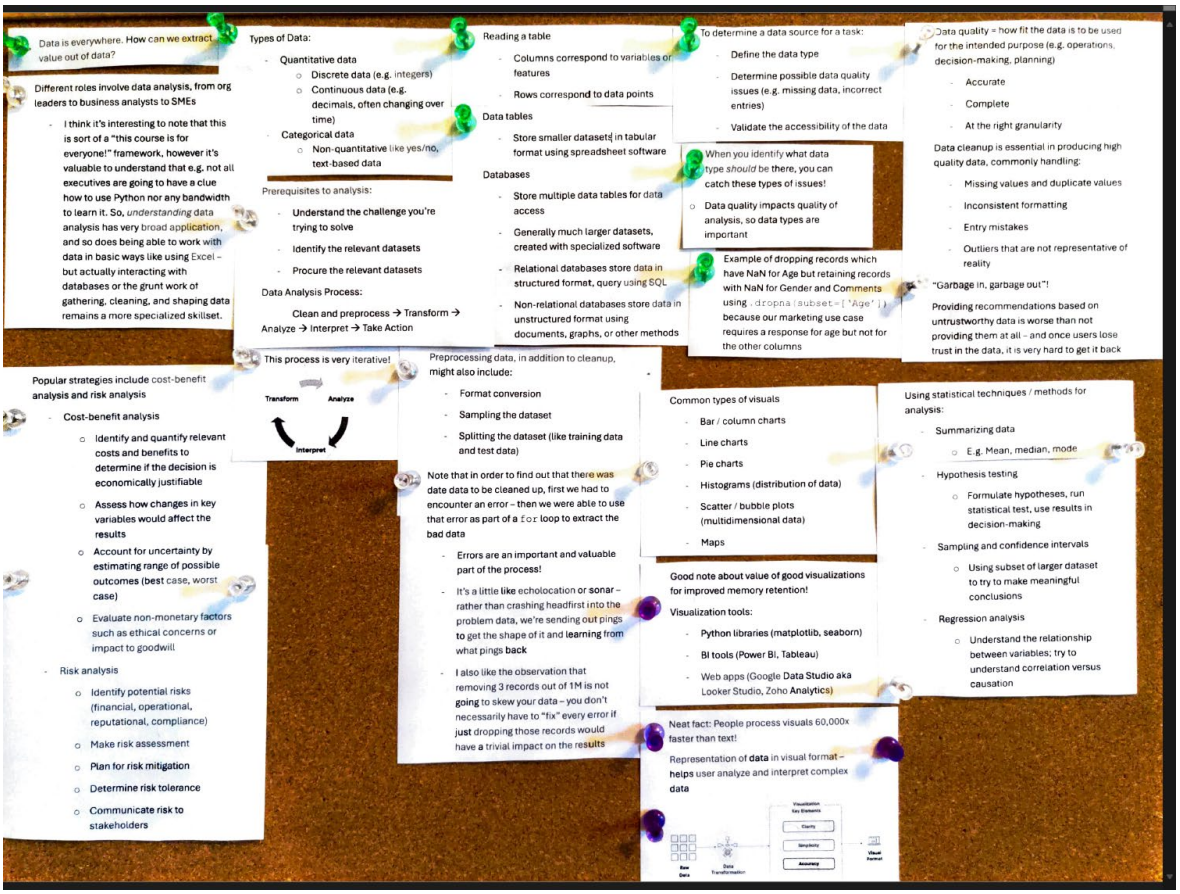
- **Data** is everywhere
  - ◊ Anything that is a form of raw information
  - ◊ Often associated with tracking events or transactions, or information like weather or location or demographics…
  - ◊ …But also includes photos, email chains, books, catalogs, the pile of unorganized receipts in the center console of your car – anything from which we can extract information
  - ◊ Fun fact: The word *data* is actually plural – the singular is *datum*
- **Information** is data that makes some kind of sense
  - ◊ That pile of receipts is **data**, but once you read it, you have some **information** about how much money you've spent on gas and takeout over the last few months, where you spent it, and how you paid
- The central question of data analysis is *how can we extract value from data?*

2.A.2 Quantitative vs. categorical

- Quantitative data is data made of numbers (also called numerical data)
  - ◊ Performing mathematical or statistical operations on quantitative data will yield new, relevant information
- Categorical data represents descriptive information
  - ◊ For example, a product's model number, color, and size are all categorical values
  - ◊ Other examples include names, location or geography, demographic information, or other text-based data
- Often categorical and quantitative data are combined in the process of analysis
  - ◊ How many blue widgets did the company sell last quarter in Nevada?

2.A.3 Structured vs. unstructured data

- Structured data is organized into a coherent format
  - ◊ Databases are a form of structured data
- Unstructured data refers to data that does not follow any kind of systematic organization
  - ◊ This could include a colorful spreadsheet that arranges an assortment of different sets of information into one human-readable sheet
  - ◊ A collection of scanned documents in a data lake is another type of unstructured data, and so is that pile of receipts we talked about



2.A.4 Common sources of data

- Business systems
  - ◊ Customer Relationship Management (CRM) software
  - ◊ Point-Of-Sale (POS) retail software
  - ◊ E-Commerce sales software
  - ◊ Financial accounting systems
  - ◊ Warehouse management systems
  - ◊ Content management systems
  - ◊ HR recruitment and hiring software
- Government agencies
  - ◊ Census data and demographic information
  - ◊ Information on parks and natural resources
  - ◊ Statewide business and industry data
  - ◊ Government spending and contracting
- Universities and research institutions
  - ◊ Scientific data
  - ◊ Survey and polling data for research
  - ◊ Meta-analyses that combine multiple sources of data

2.A.5 Common formats for data storage

- Databases
  - ◊ Relational databases using SQL
  - ◊ Other types of databases like document databases or graph databases
- Spreadsheets
  - ◊ Often spreadsheet data may have originated as an extract from a database or business system
- Exported data from software applications or online services, which may be structured as .csv files, .txt files, Excel files, JSON or XML

2.A.6 Data is fallible

- Difference between humans and computers is that computers:
  - ◊ mean exactly what they say
  - ◊ do exactly what you tell them to do
    - * This is even true of genAI tools that seem to have a mind of their own! Tools like ChatGPT are essentially returning results of very complex predictive-text equations, based on their training data and through the lens of fine-tuning
- Data might be stored in computers, but it is created by (or using the instructions of) humans … and humans make mistakes
  - ◊ Data entry errors
  - ◊ Ambiguous prompts
  - ◊ Duplication of data
  - ◊ Selection bias
- Discussion: How do you think common errors might get introduced into data that you have to analyze? What kind of “bad data” might you run into?

2.A.7 Reading data tables

- A **data table** is organized by columns and rows
- In a well-formatted table, **rows** represent individual "records" or data points
  - ◊ Calls made to a customer call center
  - ◊ Users' responses to a feedback survey
  - ◊ Items in an inventory
- **Columns** represent different variables, categories, or measures
  - ◊ For customer calls, columns might include date and time, customer name, employee name, the purpose of the call, and product(s) discussed
  - ◊ User survey responses might include the date, customer name, customer market segment, and location along with the survey questions
  - ◊ An inventory will likely include product name, brand, supplier, model number, SKU or reorder number, quantity in stock, date of expected restock for out-of-stock items, and maybe other details such as when the item was first entered into stock
- Many real-world data sources are not formatted so neatly!
  - ◊ Human-created spreadsheets may be styled as a matrix or other aesthetic arrangement – easy for humans to read, tricky for analysis

2.A.8 Tabular formats

- A table is usually displayed something like this:

Date/Time Received	Caller Name	Account Name	Segment	Duration
01/02/2024 07:55:43	Helga Booth	Booth & Meeker LLC	MLF	00:12:23
01/02/2024 08:01:03	Michael Vang	Vang and Assoc SC	SLF	00:08:48
01/02/2024 08:06:21	Saad Khan	Faegre Drinker LLP	LLF	01:02:11

- Data may be stored in different ways to create that shape
  - ◊ In a CSV file, each row of the table is represented by a new line, and the columns are created by separating each value with commas

```
Date/Time Received,Caller Name,Account Name,Segment,Duration
1/2/2024 7:55,Helga Booth,Booth & Meeker LLC,MLF,0:12:23
1/2/2024 8:01,Michael Vang,Vang and Assoc SC,SLF,0:08:48
1/2/2024 8:06,Saad Khan,Faegre Drinker LLP,LLF,1:02:11
...
```

  - ◊ Data saved in a format like JSON can also be extracted to a table

```
[
  {
    "dateTimeReceived": "1/2/2024 7:55",
    "callerName": "Helga Booth",
    "accountName": "Booth & Meeker LLC",
    "segment": "MLF",
    "duration": "0:12:23"
  },
  {
    "dateTimeReceived": "1/2/2024 8:01",
    "callerName": "Michael Vang",
    "accountName": "Vang and Assoc SC",
    "segment": "SLF",
    "duration": "0:08:48"
  },
  {
    "dateTimeReceived": "1/2/2024 8:06",
    "callerName": "Saad Khan",
    "accountName": "Faegre Drinker LLP",
    "segment": "LLF",
    "duration": "1:02:11"
  }
]
```

2.A.9 What is data analysis?

- Data analysis is the process of making sense out of data
- Because data can take many forms and come from many different sources, a wide variety of tools may be used to:
  - ◊ review available data
  - ◊ shape it into a usable data set
  - ◊ analyze relationships and trends within the data
  - ◊ generate a useful, user-friendly set of results
- We'll talk more about the data analysis lifecycle in the next section

2.A.10 Data analysis roles

- Data analysts generally have responsibility for end-to-end data analysis, from discovering data sources through creating a final output tailored to a particular purpose

- But data analysis is not limited to jobs with the title “data analyst”
  - ◊ Product managers and program managers review and analyze data relevant to their responsibilities as owner of a product or project
  - ◊ Team managers analyze department performance and HR data to understand team performance and look for opportunities to improve
  - ◊ Individual contributors may perform analysis of data relevant to their area of subject matter or domain expertise
  - ◊ However, in general, people in non-analyst roles are focusing on one part of the analysis process, not complete end-to-end analysis

- Data analysis may also be performed by database specialists, data engineers, and data scientists
  - ◊ Database specialists are more typically managing data in and out of a system, but they may also engage in analysis of the data they work with
  - ◊ Data engineers are involved in the construction and management of data systems, including implementation of data governance and security, and they may also work with building analytical tools and generating analytics
  - ◊ Data scientists often work with deep analytics involving machine learning or advanced computational engines

2.A.11 Key concepts for data analysis

- Understanding data sources is only the first step
- Analysis usually involves some form of summarizing statistical information about the data (e.g. sum of sales by quarter, or average value of a customer transaction)
- Reaching logical, well-supported conclusions is also important
  - ◊ If you know that your business has some customers who buy large orders in bulk and others who only buy a few items at a time, when is it useful to provide an “average customer order” and when is it misleading?
- Again, **curiosity** is key for solid data analysis
  - ◊ What is this data telling me?
  - ◊ What questions do I have about the data? What words are being used, and what do they mean? Are there abbreviations or acronyms that are unclear?
  - ◊ What conclusions can I draw, and what is still unknown?
  - ◊ Does this output make sense?

2.A.12 Math fundamentals

- Data analysis doesn't necessarily require you to *do* a lot of math yourself, but it does involve a lot of math concepts and calculations

- There's a lot you can do starting with simple arithmetic: + - * /
  - ◊ How do you calculate a percentage?
  - ◊ How do you calculate the average of a set of numbers?

- If you would like to brush up on your math skills, there are some great free resources available:
  - ◊ Khan Academy is a free educational resource with excellent content ranging from the basics, to statistics and probability, to calculus and linear algebra
  - ◊ Brilliant is an educational app that allows you to do daily practice in various subjects including math concepts (the free version limits you to only a few lessons per day, but a little practice is better than none!)
  - ◊ W3schools has courses on statistics and data science, as well as material on other subjects we'll cover such as Excel, SQL, and Python

2.A.13 Statistics concepts

- The basics:
  - ◊ **Mean**: Another name for the **average** of a set of numbers
  - ◊ **Median**: The **middle value** in a set of numbers
  - ◊ **Mode**: The **most frequently occurring value** in a set of numbers
  - ◊ **Range**: The difference between the highest and lowest values
- A little more advanced:
  - ◊ **Variance**: A measure of the spread of data points around the mean
  - ◊ **Standard Deviation**: The square root of the variance, representing the average distance of data points from the mean
  - ◊ **Percentile**: A measure indicating the value below which a given percentage of observations in a group of observations falls
    - ∗ 25th percentile refers to the bottom 25% of results
    - ∗ 75th percentile refers to the top 25%

2.A.14 More statistics concepts

- Correlation: A measure of the strength and direction of a linear relationship between two variables
  - ◊ On a scale of -1 to 1, where 1 is a perfect positive correlation (A goes up, B goes up); -1 a perfect negative correlation (A goes up, B goes down); and 0 indicates no relationship (B might go up or down or stay the same)
- Regression Analysis: A method to model the relationship between a dependent variable and one or more independent variables

2.A.15 Correlation and causation

- In simpler terms, correlation refers to how likely two variables are to move in the same direction
  - ◊ When the temperature starts dropping, people start buying more heavy coats; when it starts getting warmer, people buy fewer heavy coats
  - ◊ Cold temperatures and buying heavy coats are **strongly correlated**
- However, just because two variables are strongly correlated, that does not mean that one change causes the other
  - ◊ Intuitively, we understand that people buy heavy coats because the weather is cold – *not* that the weather gets cold *because* people are buying heavy coats
- Confusing correlation with causation yields misleading results
  - ◊ In the summer, there is a strong correlation between total ice cream sales and total number burglaries – the more ice cream sold, the more burglaries committed
  - ◊ Is that because eating ice cream causes people to commit burglary? Or because burglars get especially hungry for ice cream? Of course not
  - ◊ The **confounding variable** in this example is weather: ice cream sales and burglaries both tend to go up in nice weather and down in bad weather
- Remember: **correlation does not equal causation!**

2.A.16 Data at work: Business Intelligence

- Often companies use the phrase **business intelligence** (BI) to mean the technologies, processes, and tools used to collect, integrate, analyze, and share business data

- Goal of BI is to support better decision-making by identifying trends, uncovering patterns, and providing insights into business performance

- BI tools and techniques help organizations transform raw data into meaningful information, enabling leaders and stakeholders to make informed and strategic decisions

2.A.17 Data security and data governance

- **Data governance** refers to the structures and processes that an organization has put in place to manage and control data

- **Data security** more specifically refers to the systems and processes around how data classified and stored to protect confidential and sensitive information, and employee training to ensure that security is maintained

- Following rules and guidelines is critical to maintaining data security and good data governance
  - ◊ Reduce redundancy within the organization by following best practices around sharing data
  - ◊ Reduce risk of exposing confidential or highly sensitive information to unauthorized individuals
    - * This has become even more important as modern workflows incorporate AI tooling and agentic AI

Section 2.B

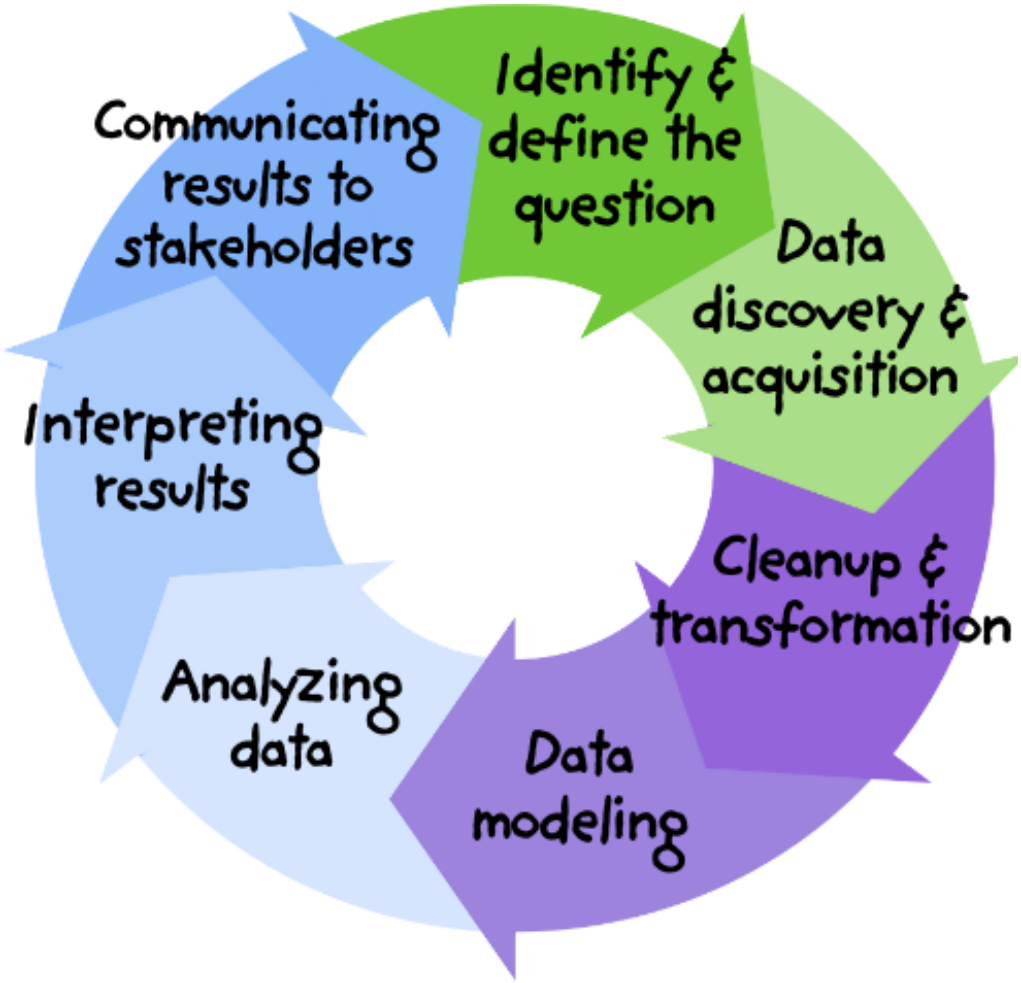
Data Analysis Lifecycle

2.B.1 Types of data analysis

- Data analysis can serve different purposes:
  - ◊ Descriptive 👀
  - ◊ Diagnostic 🩺
  - ◊ Predictive 📈
  - ◊ Prescriptive ✅
- Which of these is likely to be least intensive? Most intensive?
- How might you need to approach each of these goals differently?

2.B.2 The data analysis process

- The data analysis lifecycle includes multiple phases:
  - ◊ **Question presented** – What are we trying to understand?
  - ◊ **Data discovery and acquisition** – What data can we use? From where?
  - ◊ **Clean and transform data** – What needs to be fixed, dropped, changed?
  - ◊ **Data modeling** – How can we combine data from multiple sources?
  - ◊ **Analyze data** – What initial results can we generate from the data?
  - ◊ **Interpret analysis** – What do those results mean? What are the holes?
  - ◊ **Share results** – Who needs the results? What do they care most about?



2.B.3 Question presented

- Every data analysis project starts because someone is interested in learning something
  - ◊ How do our customers feel about our latest product release?
  - ◊ What is the projected turnover for a department over the next 5 years?
  - ◊ How many widgets did we manufacture last quarter, and how efficient was our productivity?
  - ◊ How much of our recurring subscription revenue should we attribute to each product that a subscriber either used or had access to?
- As a data analyst, the question might be coming from a different department or other stakeholder
  - ◊ People often assume that others are familiar with the same jargon or resources that they use on a regular basis
  - ◊ You may need to ask for more detail or more context
  - ◊ Even if you think you understand the initial question, it is best practice to confirm your understanding by asking clarifying questions

2.B.4 Data discovery and acquisition

- Data discovery is the process of figuring out what data you have access to and what additional data you need to address the question presented
  - For data you already have access to:
    - What is included in that data?
    - What is the format? How will you access it?
    - How comprehensive or complete is the data? How old?
    - What are the key pieces of information you expect to utilize?
  - For data you need:
    - Who might have this data? How can you request access to it?
    - If it’s a publicly available data source, where do you go to find it?
    - If applicable, what is the date range that you need?
    - As you acquire additional data, return to the discovery step: What does the new data include? How comprehensive or complete is it? Etc.

2.B.5 Clean and transform

- Raw data often contains quirky errors, incorrect data types, inconsistencies, and missing or duplicated values

- Data cleanup can include:
  - ◊ Making corrections to the data
  - ◊ Dropping bad data
  - ◊ Creating lookup tables to standardize data entries
  - ◊ Reassigning data types
  - ◊ Removing portions of the original dataset that are not relevant to the current project, such as extra columns that will not be needed

- Transformation is the process of shaping your data to suit the project you're working on:
  - ◊ Adding columns to break out specific values
  - ◊ Reshaping the data model, e.g. to ensure that relationships flow appropriately from fact tables to dimensions
  - ◊ Flattening data from related tables into one table, **or**
  - ◊ Normalizing data by separating it out into related tables
    - * We'll talk more about data normalization as we dive into databases

- There is no one-size-fits-all data cleaning process – every project will have its own unique issues and its own needs in solving them

2.B.6 Data Modeling

- Data modeling is the process of defining and organizing the relationships between your data sets so they are structured for effective analysis
  - For example, in a sales database you might link a table of transactions to a table of customer records using a shared customer ID
  - This relationship is what makes it possible to analyze sales by customer
- This can be fairly straightforward (such as when working with an existing database where table relationships are already defined) or quite complex (for instance, when combining data from multiple different sources)
  - We'll cover data modeling in more depth next week when we dive into databases, and revisit it later in the course when we work with Power BI

2.B.7 Analyze

- Once your data set is in good shape, the dive into analysis begins
- Typically analysis begins with performing basic aggregations on different dimensions of your data
- Aggregation operations include:
  - ◊ Taking the **sum** of a set of values
  - ◊ Finding the **average** or **mean** of a set of values
  - ◊ Finding the **median** or middle value
  - ◊ **Counting** categorical data
    - ∗ For instance, you might find the **sum of total sales** as a dollar amount, the **average value** of each sale, and the **count of sales** that occurred
    - ∗ Much of effective data analysis is being thoughtful and creative about how to use these straightforward operations to find insights in different slices of your data
- Other statistical analysis may also be performed
  - ◊ Regression analysis or other statistical functions
  - ◊ Running clustering or machine learning operations

2.B.8 Interpret

- Interpretation is the process of generating useful insights from the analysis
  - ◊ It is one thing to know that coats and gloves are frequently sold together, or that employee sick leave requests tend to increase before and after the Super Bowl … but what does that actually tell us?
- The interpretive process is not done in a vacuum – it is critical to understand:
  - ◊ the context of the data being reviewed and
  - ◊ the priorities of the end-user for whom the analysis is generated

2.B.9 Share results

- Analytic results may be shared in many ways
  - ◊ Will you deliver a presentation of the results to the intended audience?
  - ◊ Or will you provide a copy of a report or dashboard that end-users will review individually?
  - ◊ How important are big-picture visualizations to your audience? How important is it that they can access underlying details?
  - ◊ Can the results be re-shared by the recipients? What controls do you have in place to permit or restrict sharing?
- Sharing results does not have to be a one-way street
  - ◊ What avenue for feedback have you provided?
  - ◊ How will you address any feedback that you receive?

Module 3

Gathering Data

Section 3.A

Data Collection

3.A.1 Collecting data

- Common types of data collection:
  - ◊ Surveys
  - ◊ Customer satisfaction scores
  - ◊ User experience (UX) studies
  - ◊ Feedback forms
  - ◊ New user account setup
- Methods of collecting data:
  - ◊ Phone calls or in-person questions to individuals
  - ◊ Focus groups or panel studies
  - ◊ Fillable forms
  - ◊ Actual, physical pieces of paper with blanks to write in
  - ◊ Collaborating with e.g. customer service team to collect specific data on customer interactions

3.A.2 Good data collection practices

- Define **clear objectives**, and consider what data is **relevant** to the purpose for which it is being collected
  - ◊ Identifying information
  - ◊ Personal or demographic information
  - ◊ Contact information
  - ◊ Opinion scores
  - ◊ Comparison rankings
  - ◊ Experiences with specific products or services
- Focus on collecting data that will lead to **actionable** insights
- Use methods that ensure **consistency** in the data collected
- Set a clear **time frame** or deadline for responses to be collected
- Also consider possible **privacy** or **security** concerns
  - ◊ For example, medical provider data is subject to a very high degree of security as required by federal law
  - ◊ Personally identifiable information may also require a heightened degree of protection
    - ∗ For example, many companies will actively scrub customer service records of any data that looks like a social security number

3.A.3 Common pitfalls in data collection

- Data collection may fail to serve its purpose, or even be counterproductive, when:
  - ◊ Too much irrelevant or marginally useful information is sought, as respondents may lose interest and not finish responding
  - ◊ Data is collected that does not generate any meaningful insight into what actions ought to be taken
  - ◊ The collection method allows responses to be provided in a nonstandard format, making it hard to compare one set of responses to the next
  - ◊ The collection method is overly rigid in how responses can be provided, which can create poor quality responses and miss opportunity to generate more meaningful insights
  - ◊ No clear deadline or due date is provided, which fails to create any sense of urgency for respondents and makes it hard to determine when to cut off collection
  - ◊ Data collection is sought using a method or communication channel which yields a very low return of results
  - ◊ Private or confidential information is solicited with no clear plan on how that information will be stored or safeguarded
- Bad data collection wastes time and resources, generates unhelpful or even misleading results, and can be alienating to the audience from whom collection is sought

3.A.4 Collecting data with surveys

- Commonly used survey creation tools which are free to use, or have a free starter option:
  - Microsoft Forms, https://forms.office.com/
    - Includes several different survey templates for business purposes
    - Easily export results as an Excel spreadsheet
  - Google Forms, https://forms.google.com/
    - Templates include a very basic customer feedback form
    - Easily export results to Google Sheets
  - SurveyMonkey, https://www.surveymonkey.com/
    - Free plan includes a basic template for a customer satisfaction survey (minus the "comment box" paid feature)
    - Exporting results to Excel or .csv format requires a paid subscription
    - Can be connected to Microsoft or Google for easy sharing of results
- All of these services also include very basic summary analytics for survey responses

3.A.5 Survey basics

- Surveys might be conducted for the purpose of gathering customer feedback, market research, product research, employee satisfaction scores, etc.

- Surveys can take different formats:
  - ◊ **Online surveys** which can be sent to the desired audience via email, linked from a website, or accessed via QR code
  - ◊ **Phone surveys** such as the feedback option often provided by call center operations (e.g. “Stay on the line to answer a few brief questions”) or cold-call surveys conducted by polling firms, alumni associations, market research firms, etc.
  - ◊ **Physical forms** on actual slips of paper, which might be machine-readable or might require manual review and processing
  - ◊ **Direct interviews** or surveys completed by focus groups, where questions are asked face-to-face in a live environment

- Each of these formats has advantages and disadvantages
  - ◊ Online surveys are easy to share, and results are easy to summarize, but it can be challenging to get people to actually click through and respond, and online forms can present accessibility issues
  - ◊ Automated phone surveys may provide limited response options, making it difficult to collect nuanced data
  - ◊ Processing paper forms can be labor intensive
  - ◊ Face-to-face conversations can provide very robust information but also are typically the most time- and resource-intensive option

3.A.6 Survey best practices

- All of the "good data collection practices" apply to creating surveys, whatever format is used
- Defining clear objectives for the survey will help in writing clear, purposeful questions
  - ◊ Ask yourself, "What do I learn from the answer to this question? How does that help our objectives?"
- For each question, carefully think about the range of responses that might be given
  - ◊ You might choose to limit the available options by using multiple-choice, multi-select, or ranking questions
  - ◊ Alternatively, you might *not* want to limit the possible responses, using a text box so that respondents can type any answer
- Think about the experience of the person taking the survey, and carefully consider how much time, attention, and patience you can reasonably expect from them
- Consider when setting up your survey whether it should be anonymous, non-anonymous, or optionally anonymous
  - ◊ If you would like to be able to follow up with respondents on their answers, you will need to ask for some kind of contact information!

Section 3.B

Public Data Sources

3.B.1 Publicly available data sources

- A wide variety of data is available to the public through government agencies, nonprofit organizations, and other shared resources

- Data.gov is the central clearinghouse in the US for government data, not only including federal government agencies but also state, city or county agencies as well

- Pew Research Center is a "nonpartisan fact tank" that conducts and shares public opinion polling, demographic research, and other social science research: https://www.pewresearch.org/

- Awesome Public Datasets is a GitHub repository that provides a useful index of publicly available data, organized by topic: https://github.com/awesomedata/awesome-public-datasets

- Data Is Plural is a weekly newsletter of useful/curious datasets with a back archive of 377 editions at https://www.data-is-plural.com/
  - ◊ The archive can be exported as a spreadsheet with links to many unique and interesting data sources

3.B.2 More searchable public data sets

- AWS (Amazon's cloud service) maintains a public data exchange at https://aws.amazon.com/opendata/

- Google offers a search tool to locate public data sets at https://datasetsearch.research.google.com/

- Tableau Public is a free platform where data analysts can share data visualization projects along with the underlying data, at https://public.tableau.com/app/discover
  - ◊ Tableau is a tool for analysis and visualization somewhat similar to Microsoft Power BI
  - ◊ While we will not be using Tableau in this course, the sample reports shared on Tableau Public can be a great source of inspiration, and you can download the source data for any Tableau Public report

3.B.3 Common issues with public data

- There's no "standard" format, so public data sets tend to be extremely inconsistent from one source to the next in how they may be organized
  - ◊ This is one reason those lifecycle steps of *data discovery*, *cleaning*, and *transformation* are so important
- Some sources may be outdated, or may only capture a brief window of time
- While in general government data sources are quite reliable, other sources may be worth careful investigation – don't assume that just because a data set is publicly available, it's necessarily trustworthy
  - ◊ As with any factual information, carefully consider the source!

Section 3.C

Bias in Data

3.C.1 Bias in data collection and analysis

- **Bias** has several meanings relevant to data analysis:
  - ◊ *noun:* (1) a leaning of temperament or outlook, especially a personal and sometimes **unreasoned judgment or prejudice**; (2) an instance of such unreasoned judgment or prejudice; (3) **inclination, bent, or tendency**; (4) **systematic error introduced into sampling or testing** by selecting or encouraging one outcome or answer over others
  - ◊ *verb:* to give a slanted or prejudiced outlook to
    - ∗ example: the study's format *biases* the data in favor of large cities
- Bias can impact thinking and decision making in any area of life and business, and that includes collecting and analyzing data

3.C.2 Cognitive bias vs. cultural bias

- There are two major categories of bias that impact our thinking
  - ◊ **Cognitive bias**, shortcuts or distortions in the way our brains process certain types of problems
    - ∗ Examples include loss aversion, anchoring, and the sunk cost fallacy
  - ◊ **Cultural bias**, including biases regarding race, national origin, gender, sexuality, religion, ability/disability, and many others
- These forms of bias may operate separately or together to create erroneous thought patterns and lead people to jump to irrational conclusions
- Biases are like icebergs – there might be part that we see, but they usually go a lot deeper than we realize

3.C.3 How can data be biased?

- Bias enters into data in the human choices made in
  - how data is collected – including what questions are asked and what method of collection is used
  - from whom data is collected
  - assumptions made in how data is analyzed
  - how the results of analysis are interpreted
- Bias is often unintentional or unconscious
  - Even data that is collected by highly reputable organizations and shared with the very best of intentions can be subject to bias, simply because there was another perspective that the data failed to include
- Machine-generated data can be just as susceptible to bias as human-generated data – since, after all, humans created the machines' programming!

3.C.4 Selection bias

- In the realm of research and analytics, one common form of bias is known as **selection bias**

- Selection bias refers to a systematic error introduced into research or analysis when the sample of data is not representative of the population being studied, leading to skewed or inaccurate conclusions

- For example, studies of athletic performance and interventions to improve performance typically focus on people who are already fit
  - ◊ Preexisting fitness is a form of selection bias in these studies, and the results may not translate well to the wider population

- Another well-known example of selection bias was in the training data used to develop early facial recognition software
  - ◊ Tech companies in the US utilized sample data to train their models which included a disproportionate majority of white male faces
  - ◊ When facial recognition began to roll out commercially, it was found to do a much poorer job recognizing the faces of women and people of color (and particularly women of color)

3.C.5 Noticing and addressing bias

- To spot bias in data, consider *how, when, where,* and *from whom* the data was collected … and *by whom* it was collected
  - ◊ What are the limitations of the source data? What does it *not* include?
  - ◊ What were the interests of those collecting and organizing the data? How might that have affected the choices they made in what to include / exclude?

3.C.6 How to talk about bias in data

- You don't always have to use the word "bias" to call it out, and many people react defensively in response to that specific word
  - ◊ This can derail the conversation into whether the person "really" is biased, whether the comment "really" signifies bias, etc.
  - ◊ The "you've got poop on your shoe" rule:
    - ∗ If you can see that someone has poop on their shoe, and it stinks, it may be helpful to both of you to point it out so they can clean it up
    - ∗ But you're not obligated to point it out, especially if you think they're going to stomp around saying "How can you embarrass me like this!" or "That's not poop, it can't be, I would have noticed already!"
    - ∗ If someone points out that *you* have poop on *your* shoe, **clean it up**
- When raising issues of potential bias *in the data*, it is usually easier to have a neutral, constructive conversation *about the data*
  - ◊ Pointing out an issue in the data itself is less likely to be taken personally (again, *usually*)
  - ◊ In these situations, who/what/where language can be helpful in framing the factual issues that are the cause for concern

Module 4

Excel Fundamentals

Section 4.A

Excel Basics

4.A.1 What is Microsoft Excel?

- At its core, Excel is a software program for creating and managing spreadsheets
  - ◊ A spreadsheet is a tabular arrangement of data using columns and rows, which allows for review and summarization of that data
- People also use Excel for lots of other things – creation of calendars, project planning dashboards, info directories, etc. – but we will focus on using Excel to collect and analyze data
- Excel remains one of the most commonly adopted tools across companies worldwide
  - ◊ Similar functionality exists in other spreadsheet utilities such as the cloud-based Google Sheets and the tracking-and-automation-oriented Smartsheet, but Excel continues to be best in class for analysis

4.A.2 Navigating the Excel interface

- An Excel file is known as a **workbook** which can contain multiple **sheets** (indicated by the tabs at the bottom of the display)

- Spreadsheet data is populated into columns and rows, and each cell can be identified by its column letter and row number

- Here are a few helpful keyboard shortcuts to get familiar with:
  - ◊ Using the **tab** key will jump to the next cell horizontally to the right across a row, and shift+tab will move to the left
  - ◊ Using the **enter** key will jump down to the next row, and shift+return will jump up to the row above
  - ◊ The **arrow**, **page up / page down**, and **home** keys can all be used to navigate around the spreadsheet as well
  - ◊ Holding **shift plus an arrow key** will select multiple consecutive cells
  - ◊ Holding **control plus an arrow key** will jump to the last cell in the row or column (or to the last consecutive cell containing data or next cell after a gap in data)
  - ◊ Holding **shift+control+arrow** is a quick way to select all cells in a row or column (may need to click arrow multiple times if there are gaps)
  - ◊ When entering text in a cell, if you would like to add a line break, use **alt+return** to start text on the next line inside the cell

4.A.3 Entering and Reviewing Data

- Cells in a spreadsheet display **values** which can be numbers, text, dollar amounts, percentages, or other types of values

- To enter new value into a cell, you can:
  - ◊ Select the cell and just start typing to overwrite the cell
  - ◊ Double click the cell to enable editing within the cell
  - ◊ Select a cell and then type or edit within the formula bar at the top

- To review how a value was entered, select the cell and look at formula bar
  - ◊ When you select a cell, it will display the **evaluated value** while the formula bar will display **how that value is being generated**, whether that's a number or piece of text that was entered directly or a formula that's being used to calculate the value

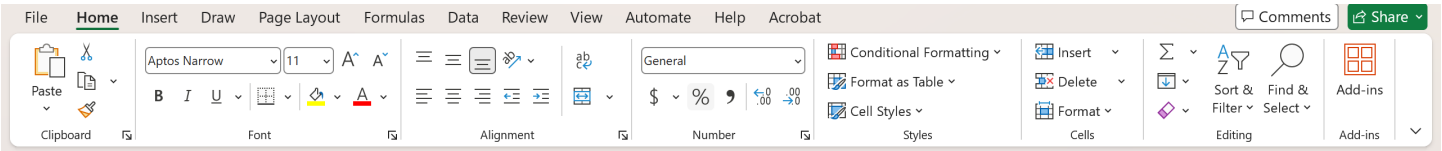
F3 =IF(VLOOKUP(E3,A:C,3,FALSE)=TRUE,"Pass","Fail")

	A	B	C	D	E	F	G	H	I
1	Name	Attempts	Pass						
2	Ana	1	TRUE		Student	Pass/Fail			
3	Bo	2	TRUE		Bo	Pass			
4	Casey	3	FALSE						
5	Dan	1	TRUE						
6	Elena	2	TRUE						

- Quick statistics: When a column or range of cells is selected, the status bar at the bottom of Excel will show available statistics (e.g. count, sum, average) for that data

4.A.4 Using the Ribbon

- Many helpful tools are available in the various tabs of the **ribbon**



- Here are just a few key tools to get familiar with:

  - **Home**: In addition to basic visual formatting, this tab includes options to change the **data type** and **number formatting** of cells, to apply **conditional formatting**, and to apply **Sort & Filter** functionality

  - **Insert**: Options on this tab allow you to create a **PivotTable** or a **chart** from your data, or to insert other features

  - **Data**: This tab includes options to **split data** into multiple columns, **remove duplicates** in data, and to **group/ungroup** multiple rows or columns

  - **Review**: Includes options to manage and navigate **comments** and to apply protection against accidental changes to the workbook

  - **View**: Includes the option to **freeze panes** to make column headers or row labels to be visible even when a user scrolls down or across the page

  - **Help**: Provides quick links to learning documentation and information from Microsoft

4.A.5 Columns and Rows

- Working with columns and rows is fundamental to organizing your data in Excel
  - ◊ Columns are vertical and identified by letters (A, B, C, etc.)
  - ◊ Rows are horizontal and identified by numbers (1, 2, 3, etc.)
  - ◊ This means every cell can be identified by its column + row (e.g. A1, B2…)
- To select an entire column or row, click on the letter or number that identifies it
  - ◊ To select multiple consecutive columns/rows, simplly click and drag
  - ◊ You can also use keyboard shortcuts – **shift** to select all consecutive items from the starting point to the next click, or **control** to select multiple individual items
- By default, values entered in a cell display on one line
  - ◊ To turn on text-wrapping, use the icon in the ribbon, or right-click and choose Format Cells, then check the Wrap Text option under Alignment

4.A.6 Working with Columns and Rows

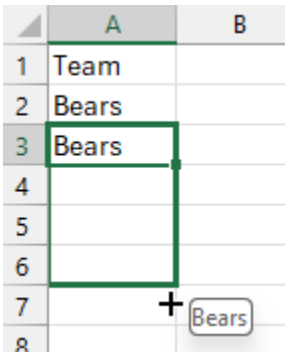
- Inserting columns or rows:
  - ◊ Right-click on a column letter or row number and select "Insert"
  - ◊ Excel will insert a new column to the left of your selection, or a new row above your selection
  - ◊ You can insert multiple columns or rows at once by selecting multiple columns/rows before right-clicking
- Deleting columns or rows:
  - ◊ Right-click on the column letter or row number and select "Delete"
  - ◊ Be careful! This will permanently remove the data in that column or row
- Adjusting column width or row height:
  - ◊ Hover your mouse over the border between two column letters (or row numbers) until you see a double-arrow cursor
  - ◊ Click and drag to resize *or* double-click to auto-fit the column width to the widest content
- Hiding columns or rows:
  - ◊ Right-click on a column letter or row number and select "Hide"
  - ◊ This doesn't delete the data – it just temporarily hides it from view
  - ◊ To unhide, select the columns or rows on either side, right-click, and select "Unhide"

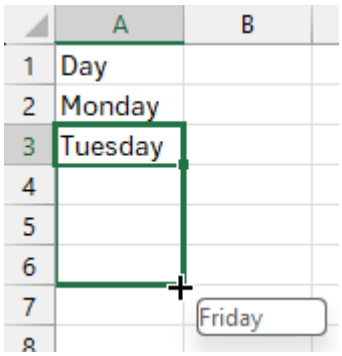
4.A.7 Ranges

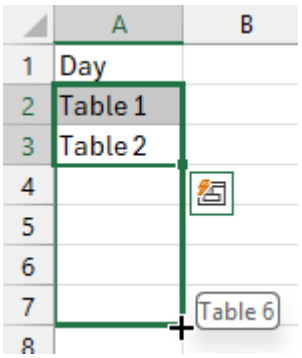
- A range is a group of one or more cells in Excel that you want to work with together
  - ◊ Ranges are written using the first cell and last cell, separated by a colon
  - ◊ For example, A1:A10 refers to cells A1 through A10 in column A
  - ◊ B2:D5 refers to a rectangular block of cells from B2 to D5
- Why ranges matter:
  - ◊ Many Excel formulas work on ranges rather than single cells
  - ◊ You can apply formatting to an entire range at once
  - ◊ Ranges make it easy to reference groups of data in calculations
- Named ranges:
  - ◊ You can give a range a custom name to make your formulas easier to read
  - ◊ Select your range, then type a name in the Name Box (to the left of the formula bar)
  - ◊ For example, you might name the range C2:C50 as "SalesData"
  - ◊ Then you can use formulas like =SUM(SalesData) instead of =SUM(C2:C50)

4.A.8 Drag to Fill and AutoFill

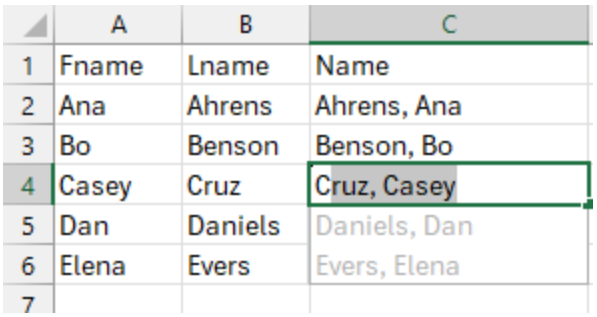
- Clicking with the + cursor at the bottom right corner of a cell enables drag-to-fill







- Where Excel detects a pattern in data that you are entering, you may also be given the option to Flash Fill column data, where predicted cell values appear in light gray



	A	B	C
1	Fname	Lname	Name
2	Ana	Ahrens	Ahrens, Ana
3	Bo	Benson	Benson, Bo
4	Casey	Cruz	Cruz, Casey
5	Dan	Daniels	Daniels, Dan
6	Elena	Evers	Evers, Elena
7			

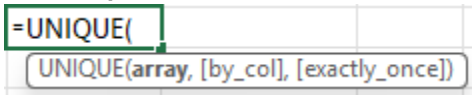
Hitting enter will populate the predicted values down the column

4.A.9 Formula Basics

- Excel formulas always begin with =

- Formulas can be used to make mathematical calculations, or to bring the value of a cell or range from one part of your workbook to another

- Formulas can also make use of **functions** to perform different types of operations
  - Examples:
    - =SUM() adds all values in a given range
    - =AVERAGE() provides an average of values in a given range
    - =COUNT provides a count of how many numbers are in a range, while =COUNTA provides a count of all values in the range (not just numbers)
    - =UNIQUE returns all the unique (non-duplicate) values in a range
  - Functions need to be provided with **arguments** (or parameters) that supply what values the function should work with and any other requirements of the function
    - The tooltip pop-up tells you what arguments the function requires and any option arguments in brackets
    - Example:

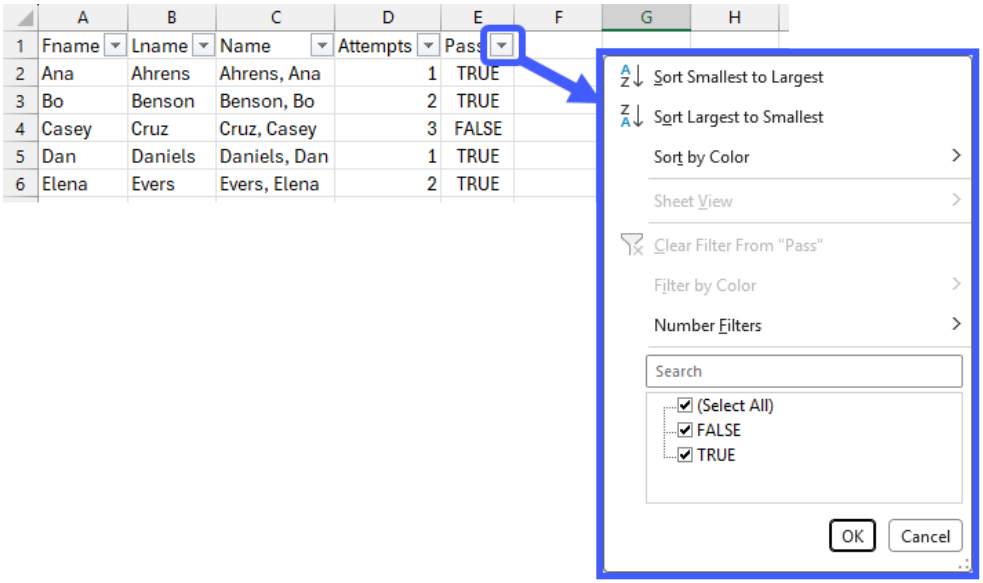


- Microsoft Learn: [Excel functions (by category)]

4.A.10 Sorting and Filtering

- Sort and filter are used for organizing and exploring your data
  - ◊ **Sorting** arranges your data in a specified sequential order (ascending or descending, whether alphabetically, numerically, by date, etc.)
  - ◊ **Filtering** temporarily hides rows that don't meet certain criteria so you can focus on the data you want to see

- To turn on Sort & Filter options:
  - ◊ Your data should be neatly organized into columns and rows, as sort/filter options will apply by column header
  - ◊ Click on any cell in the area of your data, then in the Home tab of the ribbon, select the button for Sort & Filter options, then click Filter
  - ◊ Small dropdown arrows will appear in the header row of each column
  - ◊ Click a dropdown arrow to see available options



- Sort and filter options are also available from the Data tab – try experimenting with the different tools to check out how they work

Section 4.B

Cleaning & Reshaping Data with Excel

4.B.1 Data Types and Formatting

- **Data type** refers to the kind of value, like number or text, and determines what options are available for formatting or applying calculations

- Excel automatically tries to detect what type of data you're entering, but sometimes it gets it wrong – and that can cause problems with calculations and sorting

- Common data types:
  - ◊ **General**: The default format; Excel will try to interpret what you enter
  - ◊ **Number**: Numeric values that can be used in calculations
  - ◊ **Currency**: Numbers formatted with currency symbols (like $)
  - ◊ **Date**: Dates that Excel stores as numbers behind the scenes
  - ◊ **Text**: Forces value to be treated as text (even if it looks like a number)
  - ◊ **Percentage**: Numbers displayed as percentages (0.5 becomes 50%)

- Why data types matter:
  - ◊ A date stored as text won't sort chronologically
  - ◊ A number stored as text won't calculate properly in formulas
  - ◊ An inventory number like "001" will display as "1" unless formatted as text

4.B.2 Modifying Data Types

- How to change data type:
  - ◊ Select the cell or range you want to change
  - ◊ From the home tab of the ribbon, use the Number Format dropdown (it might say "General") to select the appropriate format
  - ◊ Different format options for each data type are available by going to the Number Format dropdown and selecting More Number Formats

- Tips for specific data types:
  - ◊ Number formatting:
    - ∗ Once a cell is formatted as a number, you can adjust how many decimal places to show using the buttons
    - ∗ You can also display as a currency amount, as a percentage, or using thousand separators (commas) with the options $ % ,
  - ◊ Date formatting:
    - ∗ The default date format for full dates is month/date/year, e.g. 1/5/2026 but this can be changed using More Number Formats
    - ∗ In addition to pre-created format options under Date, you can tailor the format using Custom – e.g. the custom format `dddd` will cause the date value to display as the name of the day, e.g. Monday

- Troubleshooting tip: If you see a small green triangle in the corner of a cell, Excel is warning you that the data type might not match the content – click the cell to see options for fixing it

4.B.3 Transposing Columns and Rows

- **Transposing** means switching your rows to columns and your columns to rows
  - ◊ Example: If you have months as column headers (Jan, Feb, Mar…) and product names in rows, transposing would put months in rows and products in columns
  - ◊ This is useful when you realize your data would be easier to work with if it were organized differently

- When you might need to transpose:
  - ◊ Your data is laid out horizontally but you need it vertically
  - ◊ Column headers would make more sense as row labels (or vice versa)
  - ◊ You want to flip your data to create a different kind of chart or summary

- How to transpose data:
  - ◊ Select the range of cells you want to transpose
  - ◊ Copy the selection (Ctrl+C or right-click and choose Copy)
  - ◊ Click on the cell where you want the transposed data to start
  - ◊ Right-click and choose the clipboard icon with the arrows
    - * Make sure you have enough empty space for your transposed data – if you're transposing 5 columns and 20 rows, you'll need 20 columns and 5 rows of empty space
  - ◊ Your transposed values will paste with rows changed to columns and/or columns changed to rows

4.B.4 Removing Duplicates

- Duplicate records can skew your analysis, inflating totals or making it appear that an event occurred more frequently than it actually did

- **Remove Duplicates** helps you clean up your data by deleting duplicate rows

- When you might need to remove duplicates:
  - ◊ You want to create a list of unique values (like a list of all customers who made at least one purchase)
  - ◊ You've combined data from multiple sources, or a system error occurred, and the same records appear more than once

- How to remove duplicates:
  - ◊ Select the range of data you want to check (or click any cell in your data if it's a continuous table)
  - ◊ Go to the Data tab and click Remove Duplicates
  - ◊ Excel will show you a dialog with checkboxes for each column – check *only* the columns that should be used to identify duplicates
    - ∗ For example, you might want to de-duplicate customer records on “customer ID” but *not* “first name,” “last name,” or “state”
  - ◊ Click OK
  - ◊ Excel will tell you how many duplicate values were found and removed, and how many unique values remain

4.B.5 Cautions for Removing Duplicates

- Remove Duplicates permanently deletes de-duplicated rows
  - ◊ If you want to keep working with the original data for other purposes, consider copying the data to a new sheet and then de-duplicating, or otherwise creating a backup copy
- Excel keeps the first occurrence of each duplicate and deletes the rest
  - ◊ You may want to sort your data by another column first where there is a desired value to ensure you keep what you want
- Make sure "My data has headers" is checked if your first row contains column names
- Be thoughtful about which columns to include
  - ◊ Sometimes what looks like a duplicate is actually a separate valid record
  - ◊ Example: If you're analyzing sales data, two rows with the same Order ID are probably duplicates, but two rows with the same Customer ID might represent two different orders from the same customer

4.B.6 Splitting Cell Text to Columns

- Sometimes data is stored in a way that combines multiple pieces of information in one cell

- The **Text to Columns** tool allows you to split this combined data into separate columns

- Common scenarios:
  - ◊ Full names that you need to split into first name and last name
  - ◊ Addresses that include street, city, and state all in one cell
  - ◊ Survey data with “check all that apply” responses separated by commas
  - ◊ Dates formatted as text that need to be separated into month, day, and year

- Important notes:
  - ◊ Make sure you have empty columns to the right of your selected column, as any existing data there will be overwritten
  - ◊ The original column will be replaced with the first piece of split data
  - ◊ If splitting creates more or fewer columns than you expected, you may need to adjust your delimiter choice
    - ∗ Example: If you have "Smith, John Jacob" in one cell, using the command *and* space as delimiter will split it into three columns: “Smith”, “John”, and “Jacob” – while using just the comma as delimiter will split it into two columns: "Smith" and "John Jacob"

4.B.7 Using the Text to Columns Wizard

- How to split text to columns:
  - ◊ Select the cell(s) or full column containing the data you want to split
  - ◊ Go to the Data tab and click the button for Text to Columns

  - ◊ Choose between two options:
    - ∗ Delimited: Data is separated by a specific character (comma, space, tab, etc.) – most common
    - ∗ Fixed width: Data is organized in columns with specific character positions – less common but does sometimes come in handy
  - ◊ Follow the remaining wizard steps:
    - ∗ For delimited data, select which delimiter(s) to use – you can choose more than one, but you can only specify one character in Other
    - ∗ For fixed width data, click to select the break lines where current data should be split
    - ∗ For either option, the Data Preview panel shows what your split results will look like
  - ◊ Click Finish

4.B.8 Find and Replace

- The Find and Replace tool helps you quickly locate specific data in your workbook and optionally replace it with something else
- This is especially useful for fixing repeated errors or updating terminology across a large dataset
- Common uses:
  - ◊ Fixing misspellings (replacing "recieve" with "receive")
  - ◊ Standardizing data (replacing "NY" with "New York")
  - ◊ Updating product names or codes
  - ◊ Removing unwanted characters or spaces

4.B.9 Using Find and Replace

- How to use Find:
  - ◊ Press Ctrl+F to open the Find dialog, then type what you're looking for in the search box
  - ◊ Click Find Next to jump to each occurrence one at a time
  - ◊ Or click Find All to see a list of every occurrence
- How to use Replace:
  - ◊ Press Ctrl+H to open the Find and Replace dialog, then enter what you want to find in the "Find what" box
  - ◊ Enter what you want to replace it with in the "Replace with" box
    - ∗ If you want to delete the target text, leave “Replace with” empty
  - ◊ Click Replace to replace one occurrence at a time (so you can review each one individually)
  - ◊ Or click Replace All to replace every occurrence at once (quicker, but be mindful about possible unintended consequences! Pro tip: use Find All first to see all occurrences where the change will be made)
- Options:
  - ◊ **Match case**: Makes your search case-sensitive
  - ◊ **Match entire cell contents**: Only finds cells that contain exactly what you typed (nothing more, nothing less)
  - ◊ **Search**: Choose to search by rows or by columns
  - ◊ **Look in**: Choose to search in formulas, values, or comments

4.B.10 Helpful Functions for Data Cleanup

- There are many, *many* functions that you can use in Excel when manipulating, shaping, and analyzing data, but here are a few that frequently come in handy:
  - ◊ **IF**: helps you look for a given value in your data and provide a specific output if that value is found, or a default output if it's not found
  - ◊ **SWITCH**: similar to IF, helps you look for multiple given values and provide a specific output for each, or a default output if it is not found
  - ◊ **VLOOKUP** and **XLOOKUP**: let you cross-reference one set of data against another in Excel to populate known values based on a specified lookup value

4.B.11 Using IF and SWITCH

- `=IF(logical_test, value_if_true, [value_if_false])`
  - ◊ In plain language, IF does a logic test and returns a result:
    what condition you want to test,
    what value to return if the test passes (evaluates to TRUE),
    what value to return if the test fails (evaluates to FALSE)
  - ◊ `value_if_false` is optional, and if not specified it defaults to FALSE
  - ◊ Microsoft Learn: IF function

- `=SWITCH(expression, value1, result1, [default or value2, result2],…[default or value3, result3])`
  - ◊ SWITCH can be used to look for multiple values and return a specified result for each:
    where to search,
    what value to search for,
    what value to return if found,
    [optional next value to search for, optional value to return if found,]
    [optional default alternative value to return if not found]
  - ◊ If no alternative return value is specified, not finding a value returns #N/A
  - ◊ Microsoft Learn: SWITCH function

4.B.12 Using VLOOKUP

- `=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])`
  - ◊ In plain language, VLOOKUP runs using:
    what starting point you want to look up,
    where you want to look for it,
    the column number in the lookup range,
    [whether to return an approximate (TRUE) or exact (FALSE) match]
    - * `range_lookup` is an optional argument which defaults to TRUE, but in practice you may find yourself frequently wanting an exact match and so must specify FALSE
- VLOOKUP works left-to-right and top-down
  - ◊ In your lookup range, the value you want to look up always needs to be to the *left* and the matching value needs to be to the *right*
  - ◊ VLOOKUP searches the lookup column starting at the top and will return the first match that comes back – it will *not* identify multiple matches
- Microsoft Learn: VLOOKUP function

4.B.13 Using XLOOKUP

- Updated lookup function available in recent versions of Excel (but not in Excel 2016 or Excel 2019)

- `=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])`

- In plain language, XLOOKUP runs using:
  what starting point you want to look up,
  where you want to look for it,
  the array/range to return matches from,
  [what to return if no match is found],
  [whether to return an exact match (0), exact or next smaller (-1), exact or next larger (1), or wildcard match (2)],
  [start search from first item (1), reverse search (-1), binary by ascending (2), binary by descending (-2)]

  ◊ `[if_not_found], [match_mode], [search_mode]` are all optional

- XLOOKUP can perform a lookup across both vertical and horizontal ranges, and it's not limited to reading left-to-right

- Microsoft Learn: XLOOKUP function

4.B.14 Using Filters for Data Review

- Filters are one of the most powerful tools for exploring and reviewing your data
  - ◊ Beyond just hiding rows you don't want to see, filters help you spot patterns, identify outliers, and answer questions about your dataset

- Effective filtering strategies:
  - ◊ **Start broad, then narrow**: Filter by one criteria first, review the results, then add additional filters
  - ◊ **Look for unexpected values**: Use filters to check what unique values exist in each column – you might find data quality issues like misspellings or incorrect entries
  - ◊ **Filter by multiple criteria**: You can apply filters to multiple columns at once to drill down into specific segments of your data
  - ◊ **Get creative with number, text, and date filters**: Beyond just selecting specific values, you can filter for ranges (e.g., "greater than 100" or "dates in the last 30 days")

- Common data review tasks with filters:
  - ◊ Checking for blank cells in critical columns
  - ◊ Identifying the most frequent values or categories
  - ◊ Finding records within a specific date range
  - ◊ Spotting data that falls outside expected ranges (potential errors)
  - ◊ Reviewing subsets of data before doing calculations

Section 4.C

Analyzing and Visualizing with Excel

4.C.1 What are visualizations?

- The terms **visualization** or **visual** are general terms that refer to graphical representations of data

- There are many types of visualization:
  - ◊ Charts and graphs that show relationships or trends
  - ◊ Maps displaying fill areas or concentration of data points
  - ◊ A formatted table or matrix showing selected data
  - ◊ Infographics

- The shape of the visualization should support the story that you want to tell about the data

4.C.2 Creating Charts in Excel

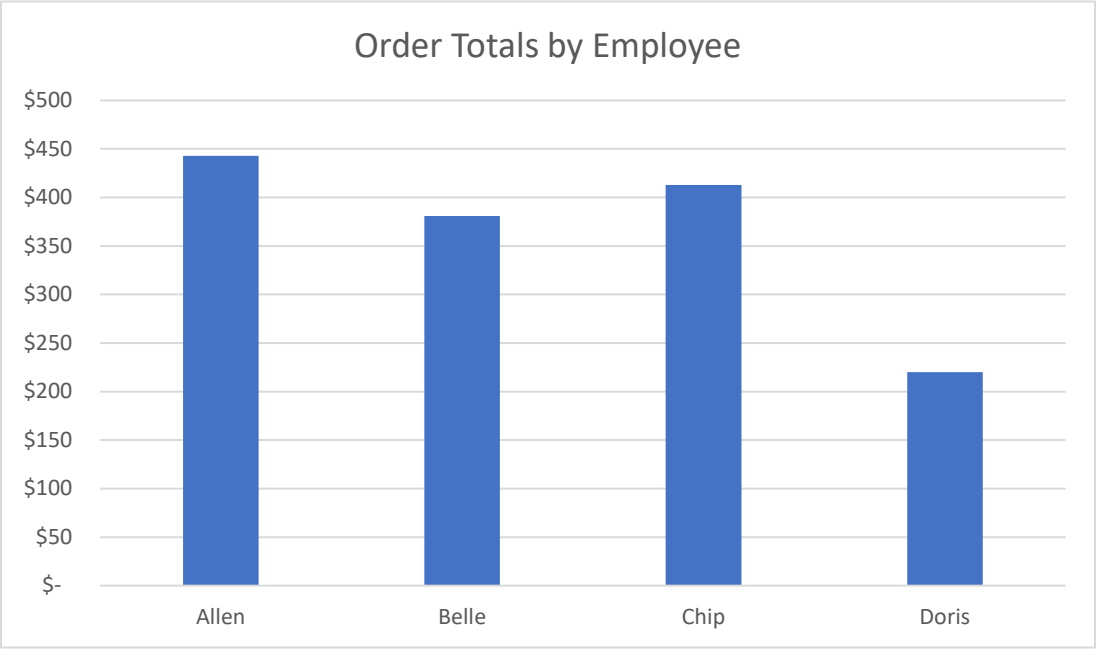
- Charts and graphs help you visualize patterns and trends in your data that might not be obvious from looking at numbers alone

- Excel makes it easy to create professional-looking charts from your data
  - ◊ Select the data you want to chart (including headers)
  - ◊ Go to the Insert tab
  - ◊ Choose a chart type from the Charts group
  - ◊ Excel will insert a basic chart that you can then customize

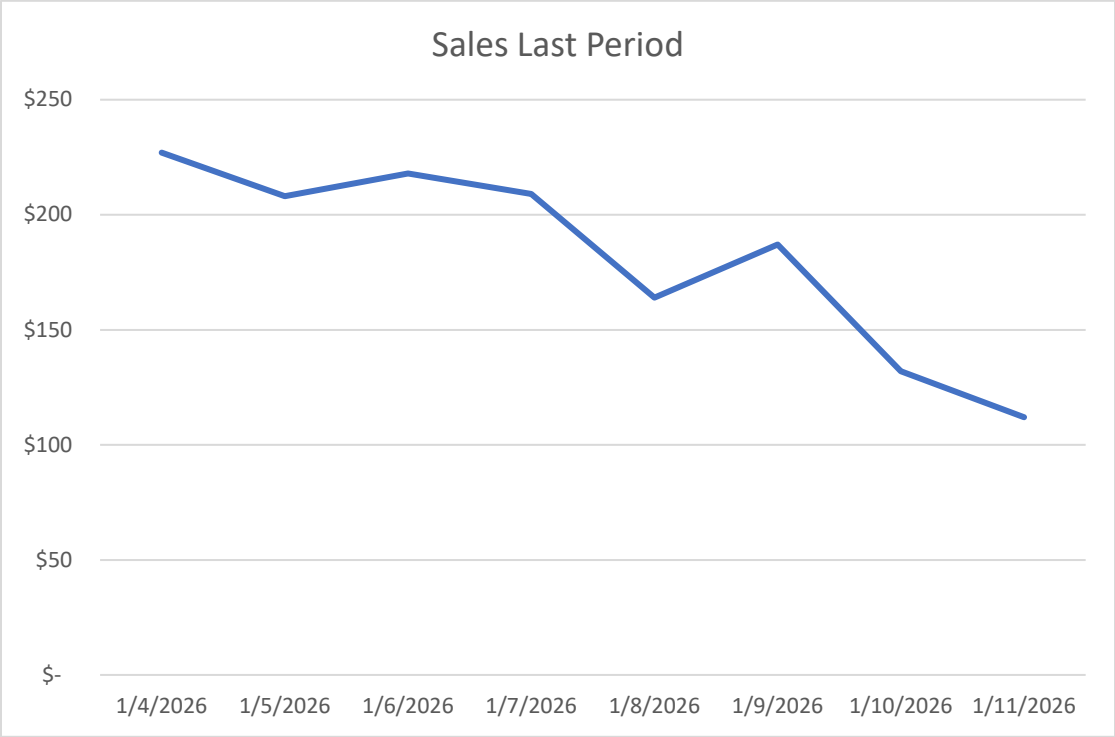
- Caution: First make sure your data is organized with clear headers and remove unnecessary rows or columns before creating a chart

4.C.3 When to Use: Bar and Line Charts

- **Bar/column charts**: Compare values across different categories

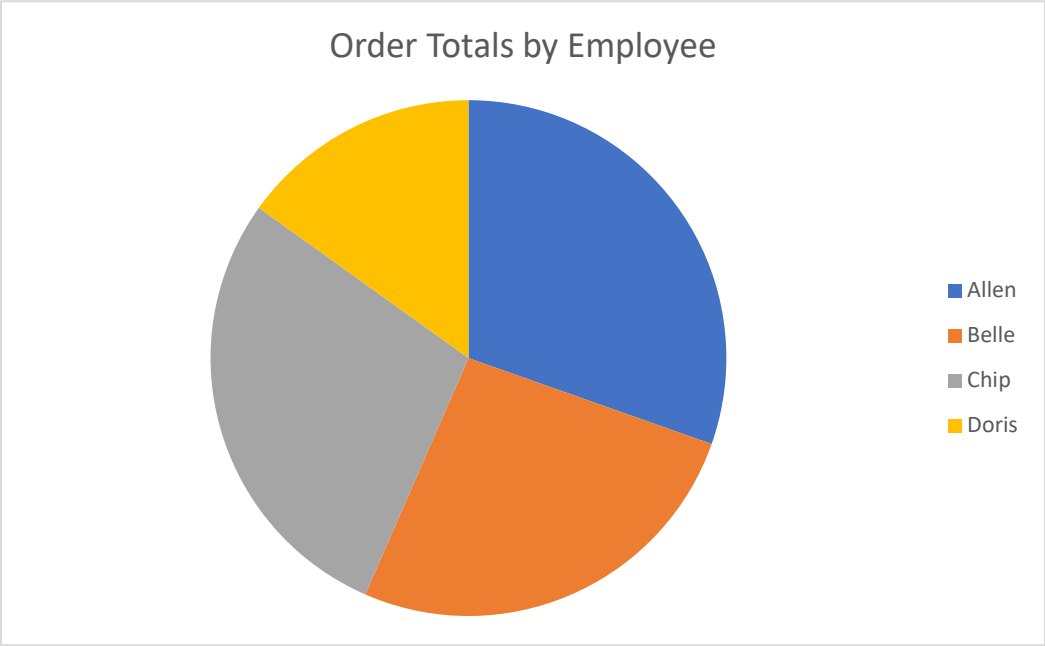


- **Line charts**: Show trends over time

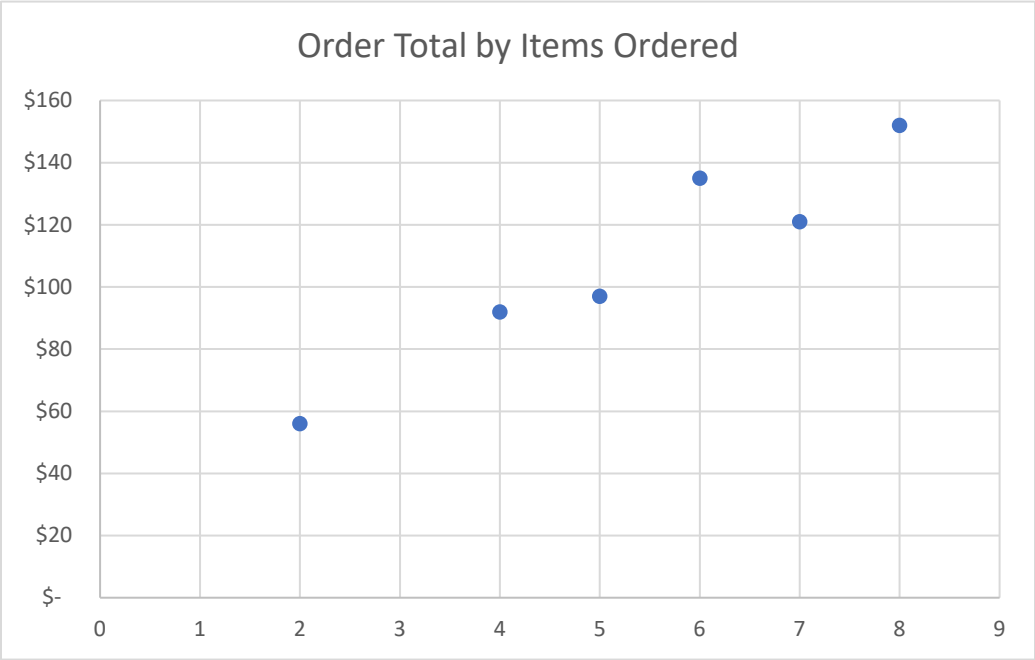


4.C.4 When to Use: Pie and Scatter Charts

- Pie charts: Show parts of a whole (best with 5-7 slices or fewer)



- Scatter plots: Show relationships between two numeric variables



4.C.5 Chart Formatting Options

- Good formatting helps your audience understand your data story quickly and clearly

- Key formatting elements:
  - ◊ Chart title: Give your chart a clear, descriptive title that explains what the data shows
  - ◊ Axis titles: Label your X and Y axes with clear, descriptive labels
  - ◊ Legend: Shows which colors or markers represent which data series
  - ◊ Data labels: Add values directly onto chart elements (bars, lines, pie slices) for easy reference
  - ◊ Gridlines: Can help with reading exact values, but use with care as too many can clutter the chart

- Accessing formatting options:
  - ◊ Click on any chart element to select it
  - ◊ Right-click and choose Format [element name] to open detailed formatting options
  - ◊ Or use the Format tab that appears when your chart is selected
  - ◊ Use the Chart Design tab to quickly change colors, styles, and layouts

4.C.6 More on Chart Formatting

- Chart styles and colors:
  - ◊ Excel offers pre-designed color schemes and styles, which are generally (but not always!) readable for colorblind users
  - ◊ If creating a custom color scheme, choose colors that have good contrast and are colorblind-friendly (have different value/saturation)
    - ∗ Pro tip: To test if your selected color scheme is accessible, use the Windows Settings option for “Turn color filters on or off” to turn your display to grayscale – if your chart becomes unreadable, then try a different color scheme

- Common formatting improvements:
  - ◊ Remove unnecessary gridlines for a cleaner look
  - ◊ Remove the legend where only a single value is displayed
  - ◊ Adjust axis scales to better show the range of your data
  - ◊ Add a descriptive subtitle or data source note
  - ◊ Increase font sizes for better readability in presentations

4.C.7 PivotTable Fundamentals

- PivotTables are one of Excel's most powerful features for analyzing and summarizing large amounts of data, letting you quickly reorganize and aggregate data with a few clicks
  - ◊ You can count, sum, average, or perform other calculations on your data
  - ◊ You can easily rearrange your analysis to look at the data from different angles
- When you might use a PivotTable:
  - ◊ Summarizing sales by region, product, or time period
  - ◊ Counting how many times different categories appear in your data
  - ◊ Finding totals or averages across different groups
  - ◊ Comparing data across multiple dimensions at once
- How to create a PivotTable:
  - ◊ Click anywhere in your data table
  - ◊ Go to the Insert tab and click PivotTable
  - ◊ Choose whether to place the PivotTable in a new worksheet or existing worksheet
  - ◊ Click OK to create a blank PivotTable

4.C.8 Building Your PivotTable

- The PivotTable Fields pane appears on the right, with available fields (based on columns) at the top, and with the bottom divided into four wells:
  - ◊ Filters: Add fields here to filter your entire PivotTable
  - ◊ Columns: Fields placed here become column headers
  - ◊ Rows: Fields placed here become row labels
  - ◊ Values: Fields placed here are calculated (summed, counted, averaged, etc.)
- To work with data in your PivotTable:
  - ◊ Drag fields from the field list into one or more of the four wells to build your analysis – usually at least Rows and Values
  - ◊ Fields added to Rows will provide row labels to begin aggregating data
    - * You can click on fields in the Rows well to change how data is grouped and displayed using Field Settings
  - ◊ For fields added to Values, Excel will automatically aggregate numeric data by sum and text fields by count
    - * You can click on fields in the Values well to change how they're calculated (sum, average, count, etc.) using Field Settings
- Pro tip: PivotTables work by connecting to your original data. If your source data for the PivotTable changes, right-click the PivotTable and choose "Refresh" to update it

4.C.9 Pivot Charts

- A PivotChart is a visual representation of a PivotTable
- Just like PivotTables, PivotCharts let you dynamically change what you're looking at to explore different views of your data
- How to create a PivotChart:
  - ◊ If you already have a PivotTable, click anywhere in it
  - ◊ Go to the PivotTable Analyze tab (or Insert tab) and click PivotChart
  - ◊ Choose a chart type and click OK
  - ◊ Alternatively, when creating a new PivotTable, you can choose PivotChart instead to create both at once
- You can format PivotCharts just like regular charts (titles, colors, styles)
- You can change the chart type at any time if a different visualization would be more effective

Module 5

Data Storytelling

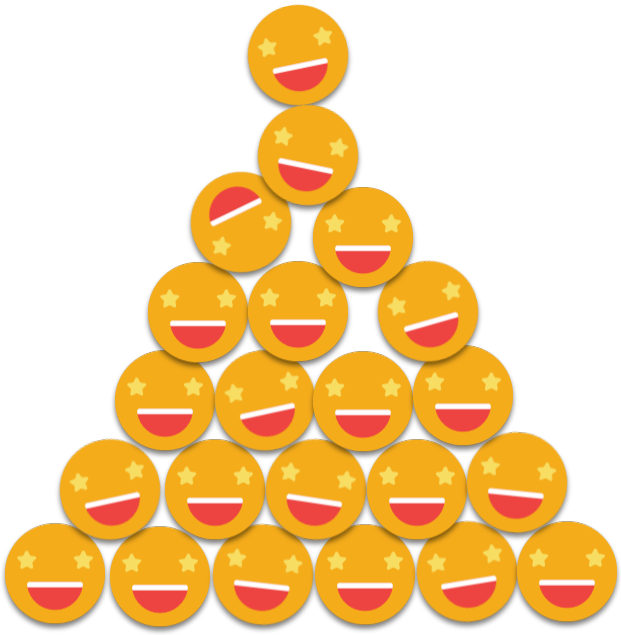
Section 5.A

Intro to Data Storytelling

5.A.1 Introduction to data storytelling

- **Data storytelling** combines narrative, visuals, and data analysis
  - ◊ Focuses on the all-important question, ***"So what?"***
  - ◊ Moves from **context** through **insights** to **action**
  - ◊ Engages your audience and brings sense and structure to complex data
- Visual storytelling drives impact
  - ◊ **Visuals are 22x more memorable** than bare facts alone

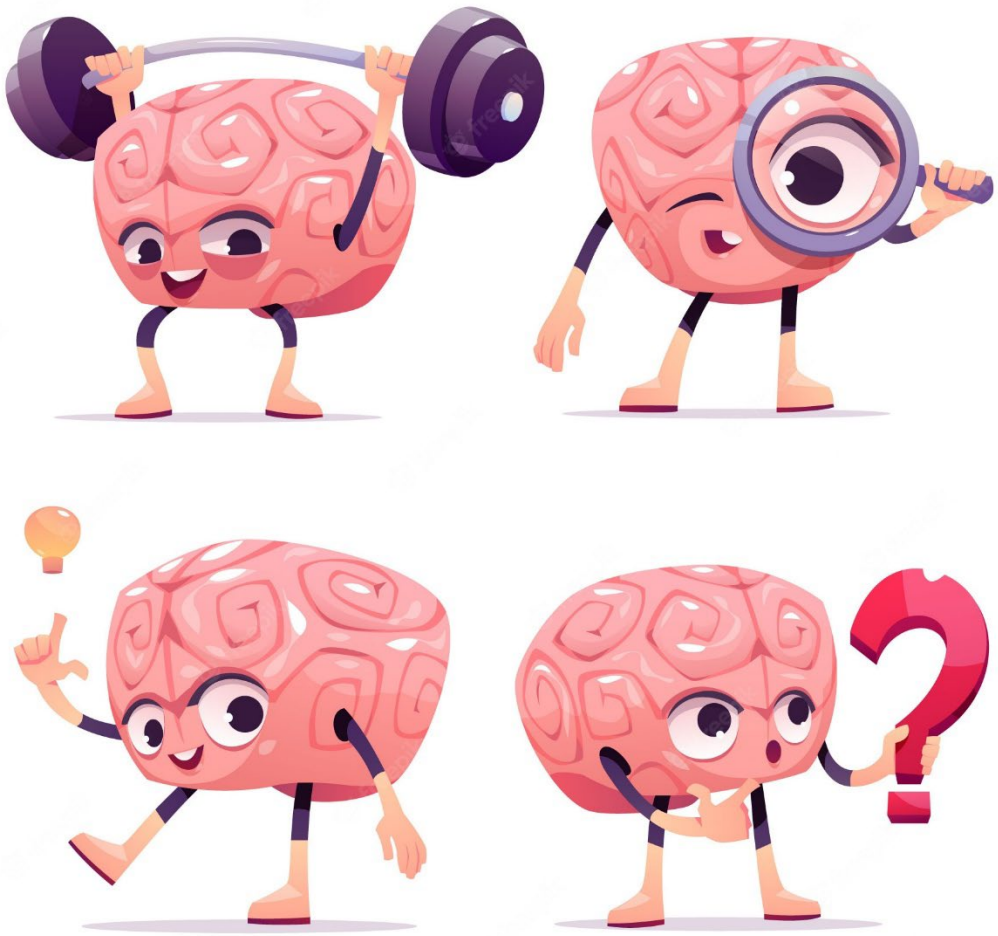
impact of the above bullet point



22x impact of this cute little visualization

5.A.2 Why does data need storytelling?

- Our brains are built around stories
  - ◊ Neural coupling allows the listener to connect the story to their own ideas and experiences
  - ◊ Dopamine is released to help remember emotionally charged stories with greater accuracy
  - ◊ Additional areas of the brain are engaged when a story evokes action and sensory perception, enhancing memorability



5.A.3 Understanding your audience

- Knowing the needs and priorities of your audience is critical to crafting an effective story

- Because data analysis is used for so many different purposes, there is no one-size-fits-all audience

- Be mindful of terminology and potentially obscure jargon when speaking to any audience, and provide definitions and context for important terms

- Think about the 5 Ws:
  - ◊ **Who** is your audience?
  - ◊ **What** are they most interested in? What are their priorities?
  - ◊ **Where** will you be sharing the results of your analysis?
  - ◊ **When** are you sharing the results? On a short or long timeframe?
  - ◊ **Why** does your audience need to know what you're sharing?

5.A.4 Structuring your data story

- Start with **clear business context**
- Focus on **key insights** over “data dumps”
- Use **appropriate visualizations**
- Provide clear **calls to action**
- Good data storytelling relies on a foundation of good data!

5.A.5 Presentation best practices

- "Presentation" includes ***any*** way that you are sharing your report
  - ◊ This may be rolling out your results in a team meeting, or via internal company site, email, Teams/Slack, or (very often!) some combination of these
  - ◊ While there are many presentation tools that you might use, the same storytelling principles apply no matter what tool you are using for analysis
- Be clear about the "So what?" up front
  - ◊ Don't assume the importance of the data speaks for itself – make the stakes clear to your audience
  - ◊ Apply the "family dinner rule" – If you were at a family dinner talking about your work, how would you explain this report to your spouse (or kid, or parent…)?
  - ◊ "One weird trick" – For a storytelling hook, try brainstorming possible titles that you might use if you were writing a click-bait version
    - * *"We found one weird trick to predict whether employees are spending their time efficiently"*
    - * *"We compared last year's hiring data to sales performance, and you'll never guess what happened next"*
    - * *"We looked at how new processes impacted production, and what we found will blow your mind"*

5.A.6 Common Pitfalls

- Analysis paralysis
  - ◊ To avoid an endless spiral of iteration, set clear milestones / deadlines and get your "good enough" analysis to the audience that needs to see it
- Misleading visualizations or misleading analysis
  - ◊ Be clear about what your data really represents! Don't manipulate visualizations for dramatic effect to the point of being misleading
- Irrelevant details or an overwhelming amount of detail
  - ◊ This is the dreaded "data dump" – make sure you focus on the *results* of your analysis, and don't get bogged down in the details of the original data or every step of the path you took to reach your results
- Failing to define key terms, overuse of jargon or abbreviations
  - ◊ Every workplace has its own internal vocabulary – just be careful that you are appropriately explaining terms at the right level for your audience
- Providing no avenue for questions or feedback
  - ◊ When presenting in a meeting, always plan to allow time for Q&A
  - ◊ To reach people with different learning/processing styles, also make sure you invite feedback another way, e.g. by asking for questions via chat or providing your email address for follow-up questions

5.A.7 Give Recommended Actions / Next Steps

- Make sure your presentation addresses that age-old question: **"So what?"**
  - ◊ Structure your shared insights to help shape decision-making
  - ◊ If it's not clear how you might do this, ask for feedback on your presentation from a supervisor or a more experienced colleague – what do they think the "So what?" story should be?
- Translate into actionable recommendations
  - ◊ Again, if you're not sure what action you should recommend, ask for feedback!
  - ◊ Sometimes making a recommendation for next steps is more about providing a starting point for discussion than a perfect road map
  - ◊ Even if the recommendation is "We need to learn more about XYZ" – at least that's a start!

End of Week 1 Student Guide